Table 2.1 shows the comparison between serial and parallel data transmission.

▼ **Table 2.1** Comparison of serial and parallel data transmission methods

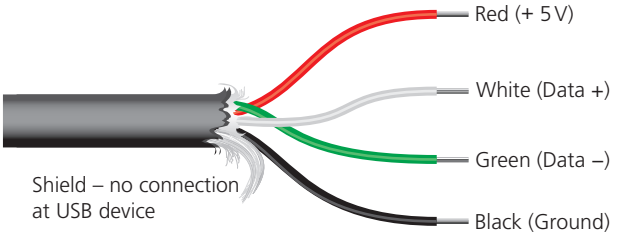
Serial	Parallel
less risk of external interference than with parallel (due to fewer wires)	faster rate of data transmission than serial
more reliable transmission over longer distances	works well over shorter distances (for example, used in internal pathways on computer circuit boards)
transmitted bits won't have the risk of being skewed (that is, out of synchronisation)	since several channels/wires used to transmit data, the bits can arrive out of synchronisation (skewed)
used if the amount of data being sent is relatively small since transmission rate is slower than parallel (for example, USB uses this method of data transmission)	preferred method when speed is important
used to send data over long distances (for example, telephone lines)	if data is time-sensitive, parallel is the most appropriate transmission method
less expensive than parallel due to fewer hardware requirements	parallel ports require more hardware, making them more expensive to implement than serial ports
	easier to program input/output operations when parallel used

2.1.3 Universal serial bus (USB)

Link

Also refer to Section 4.1 on software drivers regarding devices plugged into USB ports.

As the name suggests, the **universal serial bus (USB)** is a form of serial data transmission. USB is now the most common type of input/output port found on computers and has led to a standardisation method for the transfer of data between devices and a computer. It is important to note that USB allows both half-duplex and full-duplex data transmission.



▲ **Figure 2.12** Typical USB cable

As Figure 2.12 shows, the USB cable consists of a four-wired shielded cable, with two wires for power (red and black). The other two wires (white and green) are for data transmission. When a device is plugged into a computer using one of the USB ports:

- the computer automatically detects that a device is present (this is due to a small change in the voltage on the data signal wires in the USB cable)
- the device is automatically recognised, and the appropriate device driver software is loaded up so that the computer and device can communicate effectively
- if a new device is detected, the computer will look for the device driver that matches the device; if this is not available, the user is prompted to download the appropriate driver software (some systems do this automatically and the user will see a notice asking for permission to connect to the device website).

We will now consider the benefits and drawbacks of using the USB system:

▼ **Table 2.2** Benefits and drawbacks of USB systems

Benefits	Drawbacks
devices plugged into the computer are automatically detected and device drivers are automatically loaded up	standard USB only supports a maximum cable length of 5 m; beyond that, USB hubs are needed to extend the cable length
connections can only fit one way preventing incorrect connections being made	
it has become an industry standard, which means considerable support is available	even though USB is backward compatible, very early USB standards (V1) may not always be supported by the latest computers
can support different data transmission rates (from 1.5 Mbps to 5 Gbps)	
no need for external power source since cable supplies +5 V power	even the latest version 3 (V3) and version 4 (V4) USB-C systems have a data transfer rate which is slow compared to, for example, Ethernet connections (Note: USB V2 has a maximum data transfer rate of 480 Mbps.)
USB protocol notifies the transmitter to re- transmit data if any errors are detected; this leads to error-free data transmission	
it is relatively easy to add more USB ports if necessary, by using USB hubs	
USB is backward compatible (that is, older versions are still supported)	

A new type of USB connector, referred to as USB-C, is now becoming more common in laptops and tablets/phones. This is a 24-pin symmetrical connector which means it will fit into a USB-C port either way round. It is much smaller and thinner than older USB connectors, offers 100 watt (20 volt) power connectivity, which means full-sized devices can now be charged and it can carry data at 10 gigabits per second (10 Gbps); this means it can now support 4K video delivery.

USB-C is backward compatible (to USB 2.0 and 3.0) provided a suitable adaptor is used, and is expected to become the new industry standard (universal) format.

Activity 2.3

Ten statements are shown on the left in the table. Each of these statements is either True or False. For each statement, tick (✓) the appropriate column to indicate which statements are true and which are false.

Statement	True	False
Packets have a header which contains the IP address of the sender and the receiver		
Packets don't require any form of error checking		
USBs use a protocol that allows for error-free data transmission between device and computer		
Serial data transmission suffers from data skewing		
The longest cable length supported by USB is 5 metres or less		
Simplex data transmission occurs when data is transmitted one bit at a time		
Full-duplex data transmission involves sending 8 bits of data at a time		
USB uses serial data transfer		
Packet switching prevents loss of any data packets		
USB connections can transfer data using half-duplex or full-duplex		